The transverse z–integration of $d^3N_2^{\mathrm{NLO}}/d^3k$

UV logarithms, their finite companions, and *all* the remaining terms

Group I, Row V (the $C_1 \otimes B_2$ two–ρ row) — complete separation at $z \to x$

0. What this note delivers

The transverse integrand of Eq. (1.1) is singular when $z \to x$. The object of the exercise is to isolate that singularity. The previous note stopped after extracting the divergent logarithm; everything else was dropped. Here *nothing* is dropped. The z–integral is performed in full and split into three groups:

$$\boxed{\mathcal{W} \;\equiv\; \int d^2r\, e^{iq\cdot r}\,\frac{T(r)}{D} \;=\; \underbrace{\mathcal{W}_{\mathrm{UV}}}_{\text{the } \log(1/m^2)} \;+\; \underbrace{\mathcal{W}_{\mathrm{fin}}}_{\substack{\text{finite companion}\\ \text{of that log}}} \;+\; \underbrace{\mathcal{W}_{\mathrm{reg}}}_{\substack{\text{the eight structures that}\\ \text{were never divergent}}}}$$

The three groups are given in closed form in §8. The logic is:

1. **An exact algebraic identity** (§4) $T(r) = C_{\mathrm{UV}}(\xi)\,\dfrac{s\cdot P}{r^2} + T_{\mathrm{fin}}(r)$, valid at every r, not just asymptotically. The first piece carries the whole $z \to x$ divergence; T_{fin} is integrable at $r = 0$. $C_{\mathrm{UV}} = P_{gg}(\xi)/2N_c$.
2. **One divergent master integral** (§6), evaluated exactly — log *plus* its finite companion, with no expansion.
3. **Eight regular structures** (§7), each reduced to Bessel–function master integrals and written out.

Every step below is checked numerically (§12): the algebra symbolically against Eq. (1.1) itself, every master integral against direct two–dimensional quadrature, the assembled answer against a direct quadrature of the whole integrand, and the UV coefficient against the measured slope $d\mathcal{W}/d\log(1/m^2)$.

1. Variables, and the two typos in the source

With $r = z - x$, $s = y - x$, $P = x' - w'$, $\xi = p^+/k^+$, $\bar\xi = 1 - \xi$:

$$y - z = s - r, \qquad x - z = -r, \qquad (y-z)^2 = (r-s)^2, \qquad (y-x)^2 = s^2.$$

The longitudinal denominator and the phase are

$$p^+(y-x)^2 + (k^+ - p^+)(y-z)^2 = k^+\bar\xi D, \qquad D = (r-s)^2 + M^2, \qquad M^2 = \frac{\xi}{\bar\xi}\, s^2,$$

$$e^{-ik\cdot(w'-z)}e^{-i\xi k\cdot(z-x)} = e^{-ik\cdot w'}e^{ik\cdot x}\,e^{iq\cdot r}, \qquad q \equiv \bar\xi\, k.$$

Two abbreviations are used throughout:

$$D_0 \;\equiv\; D\big|_{r=0} \;=\; s^2 + M^2 \;=\; \frac{s^2}{\bar\xi}, \qquad\qquad q \equiv |q|.$$

The singularity of the problem sits at $r \to 0$, i.e. $z \to x$: that is where $1/r^2$ and $1/r^4$ blow up. There is a *second* place where denominators vanish, $r \to s$ i.e. $z \to y$; §3 shows it is harmless, and §10 shows what it does at $\xi \to 0$.

Two things in the source that had to be fixed before anything else.

(i) Eq. (1.8) is off by a factor 2 in one denominator. Contracting $-\frac{k^+-p^+}{p^+}\left[\frac{(y-z)^i(x-z)^m}{(x-z)^2}+\frac{(y-x)^m(y-z)^i}{2(y-x)^2}\right]$ with $\frac{P^i r^m}{P^2r^2}$ gives $\frac{(P\cdot(s-r))}{P^2r^2}\left[-1+\frac{r\cdot s}{2s^2}\right]\cdot(-\bar\xi/\xi)$, i.e.

$$\Theta_4=\frac{\bar\xi}{\xi}\,\frac{(s\cdot P)-(P\cdot r)}{P^2r^2}\left[1-\frac{r\cdot s}{\mathbf{2}\,s^2}\right],$$

not $\left[1-\frac{r\cdot s}{s^2}\right]$ as printed. This has been verified symbolically (§12A); Eqs. (1.4)–(1.7), (1.9), (1.10) are all correct as printed. The corrected Θ_4 is used below — it changes c_3, c_4, c_5 but *not* $C_{\rm UV}$.

(ii) Two conventions in (1.1) vs. (1.11). (1.1) carries a phase $e^{-ik\cdot(z-x)}$ and a prefactor p^+/k^+; (1.3) and (1.11) require instead $e^{-i\xi k\cdot(z-x)}$ (otherwise no $e^{iq\cdot r}$ survives and the z–integral has nothing to converge on at large r) and no explicit p^+/k^+. This note follows (1.3)/(1.11). The factor is overall and does not touch the z–integral; it only reweights the p^+ integral of §10, where the alternative is quoted.

2. The integrand in a nine–structure basis

Write the bracket of (1.1) as $T(r)/P^2$, so that T is the whole bracket with the universal $1/P^2$ stripped off. Contracting Eq. (1.1) structure by structure and collecting terms (all done symbolically, §12A) gives the compact statement

$$T(r)\;=\;\sum_{k=1}^{9}c_k(\xi)\,S_k(r),$$

	structure $S_k(r)$	coefficient $c_k(\xi)$	behaviour as $r\to 0$	origin
S_1	$\frac{(P\cdot r)(s\cdot r)}{r^4}$	$2\xi\bar\xi\ \ [=d_\perp\xi\bar\xi]$	$\sim 1/r^2$ — **UV**	$\Theta_1,\Theta_3,\Theta_5,\Theta_6$
S_2	$\frac{s\cdot P}{r^2}$	$\frac{\xi}{\bar\xi}+\frac{\bar\xi}{\xi}$	$\sim 1/r^2$ — **UV**	Θ_2,Θ_4
S_3	$\frac{P\cdot r}{r^2}$	$\xi^2-\frac{5}{2}\xi+2-\frac{1}{\xi}$	$\sim 1/r$ — integrable	all six
S_4	$\frac{(P\cdot r)(s\cdot r)}{s^2r^2}$	$\frac{\bar\xi^2}{2\xi}$	$O(1)$	Θ_3,Θ_4
S_5	$\frac{(s\cdot P)(s\cdot r)}{s^2r^2}$	$-\frac{1}{2\xi}$	$O(1)$	Θ_4,Θ_6
S_6	$\frac{s\cdot P}{s^2}$	$\frac{1}{2}$	$O(1)$	Θ_6
S_7	$\frac{(s\cdot P)\,r\cdot(r-s)}{r^2(r-s)^2}$	$-\frac{\xi}{2\bar\xi}$	$\sim 1/r$ — integrable	Θ_2
S_8	$\frac{(P\cdot r)\,s\cdot(r-s)}{r^2(r-s)^2}$	$\frac{\xi}{2}$	$\sim 1/r$ — integrable	Θ_5
S_9	$\frac{(s\cdot r)\,P\cdot(r-s)}{r^2(r-s)^2}$	$-\frac{1}{2}$	$\sim 1/r$ — integrable	Θ_6

Two remarks on the table. First, c_1 is the only place the transverse dimension enters: $c_1=d_\perp\xi\bar\xi$, from the instantaneous $\delta^{im}d_\perp$ term of Θ_1, and $d_\perp=2$ has been used elsewhere. (In $d_\perp=2-2\epsilon$ the $d_\perp$ of c_1 is cancelled by the $1/d$ of the angular average, $\langle\hat r^i\hat r^j\rangle=\delta^{ij}/d$, so the $\xi(1-\xi)$ of P_{gg} is exactly $\xi\bar\xi$ with no $O(\epsilon)$ correction; see the UV note.) Second, the coefficient of S_1 is *not* Θ_1 alone: Θ_1 gives $d_\perp\xi\bar\xi$, and $\Theta_3,\Theta_5,\Theta_6$ give $-2\bar\xi-2\xi+2=0$. Their cancellation is what the previous note meant by "T_3,T_5,T_6 drop out"; it is a statement about S_1 only, and those three structures certainly do *not* drop out of the rest of the answer.

3. Where the singularities are — power counting

3.1 $r \to 0$ *($z \to x$) — the UV*

Near $r = 0$ one has $D \to D_0$ and $(r-s)^2 \to s^2$, both finite and nonzero, and $e^{iq\cdot r} \to 1$. In two dimensions $d^2 r = r\,dr\,d\theta$, so a term behaving like r^{-n} is log–divergent for $n = 2$, convergent for $n \le 1$. Using $\langle \hat r^i \rangle = 0$ and $\langle \hat r^i \hat r^j \rangle = \delta^{ij}/2$:

$$\Big\langle \frac{(P\cdot r)(s\cdot r)}{r^4} \Big\rangle = \frac{s\cdot P}{2r^2}, \qquad \Big\langle \frac{s\cdot P}{r^2} \Big\rangle = \frac{s\cdot P}{r^2}, \qquad \Big\langle \frac{P\cdot r}{r^2} \Big\rangle = 0.$$

Only S_1 and S_2 survive. S_3, S_7, S_8, S_9 go like $1/r$ — odd, hence angular–averaging to zero at leading order, and in any case integrable; S_4, S_5, S_6 are bounded. So **the entire $z \to x$ divergence lives in S_1 and S_2**, with

$$\frac{c_1}{2} + c_2 \;=\; \xi\bar\xi + \frac{\xi}{\bar\xi} + \frac{\bar\xi}{\xi} \;\equiv\; C_{\rm UV}(\xi) \;=\; \frac{P_{gg}(\xi)}{2N_c}, \qquad P_{gg} = 2N_c\Big[\frac{\xi}{1-\xi} + \frac{1-\xi}{\xi} + \xi(1-\xi)\Big].$$

The one–to–one correspondence is: $\xi(1-\xi)$ from the instantaneous Θ_1, $\xi/(1-\xi)$ from Θ_2, $(1-\xi)/\xi$ from Θ_4.

3.2 $r \to s$ *($z \to y$) — not a divergence*

S_7, S_8, S_9 have $(r-s)^2$ in the denominator, but each numerator vanishes *linearly* there,

$$r\cdot(r-s)\Big|_{r\to s} = s\cdot(r-s) + O\big((r-s)^2\big), \qquad s\cdot(r-s), \qquad P\cdot(r-s),$$

so each behaves as $1/|r-s|$, which is integrable in two dimensions. $D = (r-s)^2 + M^2$ is regular there for $M^2 > 0$. Hence there is **no divergence at $z \to y$** and no second regulator is needed — unlike in the previous note, where an auxiliary μ^2 was introduced and its cancellation checked. Once $M^2 \to 0$, however ($\xi \to 0$), the $z \to y$ region does generate a $\log M^2$; this is worked out in §10.3 and, importantly, it *cancels* between the groups.

3.3 $r \to \infty$

S_6 carries no $1/r^2$ at all, so $S_6/D \sim r^{-2}$: the z–integral converges at large z only through the oscillation $e^{iq\cdot r}$. Several other terms are likewise only conditionally convergent. This is why the phase must be $e^{-i\xi k\cdot(z-x)}$ (see §1) — without $q = \bar\xi k$ the large–z end does not converge.

4. The exact UV/finite split of the integrand

The angular argument of §3.1 identifies the divergence but is asymptotic. It can be turned into an *exact identity*. Split S_1 into its trace and its traceless part,

$$\frac{(P\cdot r)(s\cdot r)}{r^4} = \underbrace{\frac{s\cdot P}{2r^2}}_{\text{trace}} + \underbrace{P_i s_j\Big[\frac{r^i r^j}{r^4} - \frac{\delta^{ij}}{2r^2}\Big]}_{\text{traceless}}.$$

The traceless bracket is orthogonal to every function of r^2 alone, so it contributes no logarithm: its z–integral converges at $r = 0$. Since $\frac{c_1}{2} + c_2 = C_{\rm UV}$, the trace piece of S_1 and all of S_2 combine into a single term and

$$T(r) \;=\; C_{\rm UV}(\xi)\,\frac{s\cdot P}{r^2} \;+\; T_{\rm fin}(r), \qquad C_{\rm UV}(\xi) = \xi\bar\xi + \frac{\xi}{\bar\xi} + \frac{\bar\xi}{\xi} = \frac{P_{gg}(\xi)}{2N_c},$$

$$\begin{aligned} T_{\rm fin}(r) = 2\xi\bar\xi\left[\frac{(P\cdot r)(s\cdot r)}{r^4} - \frac{s\cdot P}{2r^2}\right] & \qquad\qquad \text{(traceless, UV finite)} \\ + \Big(\xi^2 - \tfrac{5}{2}\xi + 2 - \tfrac{1}{\xi}\Big)\frac{P\cdot r}{r^2} + \frac{\bar\xi^2}{2\xi}\,\frac{(P\cdot r)(s\cdot r)}{s^2 r^2} - \frac{1}{2\xi}\,\frac{(s\cdot P)(s\cdot r)}{s^2 r^2} + \frac{1}{2}\,\frac{s\cdot P}{s^2} \\ - \frac{\xi}{2\bar\xi}\,\frac{(s\cdot P)\,r\cdot(r-s)}{r^2(r-s)^2} + \frac{\xi}{2}\,\frac{(P\cdot r)\,s\cdot(r-s)}{r^2(r-s)^2} - \frac{1}{2}\,\frac{(s\cdot r)\,P\cdot(r-s)}{r^2(r-s)^2}. \end{aligned}$$

This is the central algebraic statement of the note, and it is an identity in r — verified symbolically against Eq. (1.1) itself (§12A). Everything after this point is just the evaluation of two–dimensional Fourier integrals. Note that $T_{\rm fin}$ is *not* small: it contains $1/\xi$ and $1/\bar\xi$ poles of its own, and at $\xi \to 1$ it does not vanish (§10.4).

5. Master integrals

Every term of T/D is of the form $N(r)\,/\big[(r^2)^a\,(r-s)^{2b}\,D\big]$ with $a\in\{0,1,2\}$, $b\in\{0,1\}$ and $N\in\{1,r^i,r^ir^j\}$. Since $(r-s)^2$ and $D=(r-s)^2+M^2$ differ only by a constant, partial fractions

$$\frac{1}{(r-s)^2\,D}=\frac{1}{M^2}\left[\frac{1}{(r-s)^2}-\frac{1}{D}\right]$$

reduce everything to *two*–denominator integrals: the pair $\{r^2,D\}$ and the massless pair $\{r^2,(r-s)^2\}$.

5.1 Feynman parametrisation

With $A=r^2+m^2$ (m the UV regulator, needed only for the divergent master) and $B=(r-s)^2+\Delta^2$,

$$\alpha A+(1-\alpha)B=(r-b)^2+\Delta,\qquad b=\bar\alpha\,s,\qquad \Delta(\alpha)=\bar\alpha\big(\alpha s^2+\Delta^2\big)+\alpha m^2,\qquad \bar\alpha\equiv 1-\alpha,$$

$$\frac{1}{AB}=\int_0^1\frac{d\alpha}{[\alpha A+\bar\alpha B]^2},\qquad \frac{1}{A^2B}=\int_0^1\frac{2\alpha\,d\alpha}{[\alpha A+\bar\alpha B]^3}.$$

For the pair $\{r^2,D\}$ take $\Delta^2=M^2$; for the massless pair $\{r^2,(r-s)^2\}$ take $\Delta^2=m^2=0$, i.e. $\Delta_0(\alpha)=\alpha\bar\alpha s^2$.

5.2 The two–dimensional Bessel transforms

Shift $r=b+u$; the phase factorises, $e^{iq\cdot r}=E\,e^{iq\cdot u}$ with $E\equiv e^{i\bar\alpha\,q\cdot s}$. Everything then follows from the 2D Yukawa transform and two derivative rules, with $\lambda\equiv q\sqrt{\Delta}$:

$$U_1=\int\frac{d^2u\,e^{iq\cdot u}}{u^2+\Delta}=2\pi K_0(\lambda),\qquad U_{n+1}=-\frac{1}{n}\frac{\partial U_n}{\partial\Delta},\qquad V_n^i=-i\frac{\partial U_n}{\partial q_i},\qquad T_n^{ij}=-\frac{\partial^2U_n}{\partial q_i\partial q_j}.$$

Using $K_0'=-K_1$ and $K_1'=-K_0-K_1/x$, and $\frac{d}{d\Delta}\big[\sqrt{\Delta}K_1(q\sqrt{\Delta})\big]=-\frac{q}{2}K_0$:

$$U_2=\frac{\pi q}{\sqrt{\Delta}}K_1(\lambda),\qquad U_3=\frac{\pi q}{2}\left[\frac{K_1(\lambda)}{\Delta^{3/2}}+\frac{q\,K_0(\lambda)}{2\Delta}\right],$$

$$V_2^i=i\pi q^iK_0(\lambda),\qquad V_3^i=\frac{i\pi q^iq}{4\sqrt{\Delta}}K_1(\lambda),$$

$$T_2^{ij}=\pi\left[\delta^{ij}K_0(\lambda)-\frac{q^iq^j\sqrt{\Delta}}{q}K_1(\lambda)\right],\qquad T_3^{ij}=\frac{\pi}{4}\left[\frac{\delta^{ij}q}{\sqrt{\Delta}}K_1(\lambda)-q^iq^jK_0(\lambda)\right].$$

5.3 The masters

Collecting, with $b=\bar\alpha s$ and $E=e^{i\bar\alpha q\cdot s}$, the six masters that occur are (all UV finite except $\mathcal{G}$):

$$\Phi\;=\;\int d^2r\,\frac{e^{iq\cdot r}}{D}\;=\;2\pi\,e^{iq\cdot s}K_0(Mq)\qquad\text{(closed form, no parameter integral)},$$

$$\Psi^i=\int d^2r\,\frac{e^{iq\cdot r}\,r^i}{r^2D}=\pi\int_0^1 d\alpha\,E\left[\bar\alpha\,s^i\frac{q}{\sqrt{\Delta}}K_1(\lambda)+i\,q^iK_0(\lambda)\right],$$

$$\mathcal{N}^{ij}=\int d^2r\,\frac{e^{iq\cdot r}\,r^ir^j}{r^2D}=\int_0^1 d\alpha\,E\Big[b^ib^jU_2+b^iV_2^j+b^jV_2^i+T_2^{ij}\Big],$$

$$\mathcal{T}^{ij}=\int d^2r\,e^{iq\cdot r}\frac{r^ir^j-\frac12\delta^{ij}r^2}{r^4\,D}=\int_0^1 d\alpha\,E\,\Big\{2\alpha\big[b^ib^jU_3+b^iV_3^j+b^jV_3^i+T_3^{ij}\big]\Big\}_{\text{traceless}},$$

$$\mathcal{G}=\int d^2r\,\frac{e^{iq\cdot r}}{(r^2+m^2)\,D}=\int_0^1 d\alpha\,E\,U_2(\Delta),\qquad \Delta=\bar\alpha(\alpha s^2+M^2)+\alpha m^2.$$

In $\Phi,\Psi,\mathcal{N},\mathcal{T}$ one may set $m=0$, so $\Delta=\bar\alpha(\alpha s^2+M^2)$. Three checks that the reader can make immediately: $\mathcal{N}^{ii}=\Phi$ (because $r^ir^i/r^2=1$); $\mathcal{T}^{ii}=0$ by construction; and $\mathcal{T}$ is finite at $\alpha\to1$ only *after* the traceless projection — the divergent piece of $2\alpha T_3^{ij}$ is pure δ^{ij}. Taking the trace *before* projecting is the one trap here: $\int\frac{r^ir^j}{(r^2+m^2)^2D}$ has trace $\mathcal{G}-m^2\int\frac{1}{(r^2+m^2)^2D}$, and the second term does *not* vanish as $m\to0$ — it tends to π/D_0.

5.4 The massless masters

For the pair $\{r^2, (r-s)^2\}$, with $\Delta_0 = \alpha\bar\alpha s^2$, $\lambda_0 = qs\sqrt{\alpha\bar\alpha}$, $r = u + \bar\alpha s$ and $r - s = u - \alpha s$:

$$A_7 = \int d^2r\, e^{iq\cdot r}\frac{r\cdot(r-s)}{r^2(r-s)^2} = \pi\int_0^1 d\alpha\, E\Big[2K_0(\lambda_0) - 2\lambda_0 K_1(\lambda_0) + i(1-2\alpha)(q\cdot s)K_0(\lambda_0)\Big],$$

$$A_8 = \int d^2r\, e^{iq\cdot r}\frac{(P\cdot r)\, s\cdot(r-s)}{r^2(r-s)^2} = \int_0^1 d\alpha\, E\Big[P_i s_j T_2^{ij} - \alpha s^2(P\cdot V_2) + \bar\alpha(P\cdot s)(s\cdot V_2) - \alpha\bar\alpha s^2(P\cdot s)U_2\Big],$$

$$A_9 = \int d^2r\, e^{iq\cdot r}\frac{(s\cdot r)\, P\cdot(r-s)}{r^2(r-s)^2} = \int_0^1 d\alpha\, E\Big[s_i P_j T_2^{ij} - \alpha(P\cdot s)(s\cdot V_2) + \bar\alpha s^2(P\cdot V_2) - \alpha\bar\alpha s^2(P\cdot s)U_2\Big].$$

(A_7 used $u^2 = (u^2+\Delta_0) - \Delta_0$, so that $\int u^2 e^{iqu}/(u^2+\Delta_0)^2 = U_1 - \Delta_0 U_2$, and $\Delta_0 = \alpha\bar\alpha s^2$ collapses the two U_2 terms into $-2\lambda_0 K_1$.) These are ξ–independent: they depend on q and s only. A useful identity: $A_8 = A_9$ when $P = s$, since then the two numerators coincide.

6. The divergent master, exactly — log *and* companion

There is no need to expand anything. Use the identity $\frac{1}{r^2 D} = \frac{1}{r^2 D_0} + \frac{1}{r^2}\big(\frac{1}{D} - \frac{1}{D_0}\big)$ together with

$$\frac{1}{D} - \frac{1}{D_0} = \frac{D_0 - D}{D\,D_0} = \frac{2r\cdot s - r^2}{D\,D_0} \qquad\Longrightarrow\qquad \frac{1}{r^2}\Big(\frac{1}{D} - \frac{1}{D_0}\Big) = \frac{1}{D_0}\left[\frac{2\,r\cdot s}{r^2 D} - \frac{1}{D}\right],$$

which is regular at $r = 0$ (the bracket vanishes linearly there), so m may be set to zero in it. Hence, exactly,

$$\mathcal{G} = \int \frac{d^2r\; e^{iq\cdot r}}{(r^2+m^2)\,D} = \frac{2\pi}{D_0}K_0(mq) \;+\; \frac{1}{D_0}\Big[\,2\,s\cdot\Psi - \Phi\,\Big] \;+\; O(m^2),$$

$$2\pi K_0(mq) \;=\; \pi\log\frac{4e^{-2\gamma_E}}{m^2q^2} \;=\; \underbrace{\pi\log\frac{D_0}{m^2}}_{\text{the UV log}} \;+\; \underbrace{\pi\log\frac{4e^{-2\gamma_E}}{q^2 D_0}}_{\text{finite}}\,.$$

The second line is a *choice* of where to cut the logarithm; the argument $D_0 = s^2/\bar\xi$ is picked so that the UV log reads $\log\big[(y-x)^2/\bar\xi m^2\big]$, which is the combination that pairs with the running coupling. Any other choice moves a constant between $\mathcal{W}_{\rm UV}$ and $\mathcal{W}_{\rm fin}$ and changes neither separately–meaningful physics nor the sum (see §11).

> **Dimensional regularisation.** The whole m–dependence is the single factor $\pi\log(1/m^2)$. In $d_\perp = 2 + 2\epsilon_{\rm UV}$, $\int d^{d_\perp}r\, e^{iq\cdot r}/r^2 = \pi^{1+\epsilon}2^{2\epsilon}\Gamma(\epsilon)(q^2)^{-\epsilon}$, and matching the two regulators gives the dictionary
>
> $$\log\frac{1}{m^2} \;\longleftrightarrow\; \frac{1}{\epsilon_{\rm UV}} + \log\big(\pi e^{\gamma_E}\mu^2\big),$$
>
> i.e. $\pi\log(1/m^2) \to \pi/\epsilon_{\rm UV} + \ldots$, the pole that cancels against the $-2/\epsilon$ of the gluon self–energy counterterm. (With the opposite convention $d_\perp = 2 - 2\epsilon$ the pole is $-1/\epsilon$.)

7. The eight regular structures, reduced

Each surviving structure maps onto the masters of §5. The three $(r-s)^2$ structures are handled by the partial fraction of §5, which turns $\int N/[r^2(r-s)^2 D]$ into $\frac{1}{M^2}\big[\int\frac{N}{r^2(r-s)^2} - \int\frac{N}{r^2 D}\big]$:

$$\mathcal{S}_7 \equiv \int d^2r\, e^{iq\cdot r}\frac{(s\cdot P)\, r\cdot(r-s)}{r^2(r-s)^2 D} = \frac{s\cdot P}{M^2}\Big[A_7 - \Phi + s\cdot\Psi\Big],$$

$$\mathcal{S}_8 \equiv \int d^2r\, e^{iq\cdot r}\frac{(P\cdot r)\, s\cdot(r-s)}{r^2(r-s)^2 D} = \frac{1}{M^2}\Big[A_8 - P_i s_j\mathcal{N}^{ij} + s^2\,(P\cdot\Psi)\Big],$$

$$\mathcal{S}_9 \equiv \int d^2r\, e^{iq\cdot r}\frac{(s\cdot r)\, P\cdot(r-s)}{r^2(r-s)^2 D} = \frac{1}{M^2}\Big[A_9 - s_i P_j\mathcal{N}^{ij} + (P\cdot s)\,(s\cdot\Psi)\Big],$$

using $r\cdot(r-s) = r^2 - r\cdot s$, $s\cdot(r-s) = s\cdot r - s^2$, $P\cdot(r-s) = P\cdot r - P\cdot s$ and $\int e^{iq\cdot r}r^2/(r^2 D) = \Phi$. The apparent $1/M^2$ poles are spurious: at $M^2 = 0$ one has $D = (r-s)^2$ and the bracket vanishes identically, so each $\mathcal{S}_k$ is finite as

$\xi \to 0$.

structure	**coefficient**	z**–integral** $\int d^2r\, e^{iq\cdot r} S_k/D$
$S_1 - \frac{1}{2}S_2$ (traceless)	$2\xi\bar{\xi}$	$P_i s_j\, \mathcal{T}^{ij}$
S_2 (trace + S_2)	$C_{\rm UV}$	$(s\cdot P)\ \mathcal{G}$ — the only divergent one
S_3	$\xi^2 - \frac{5}{2}\xi + 2 - \frac{1}{\xi}$	$P\cdot\Psi$
S_4	$\bar{\xi}^2/2\xi$	$P_i s_j\, \mathcal{N}^{ij}/s^2$
S_5	$-1/2\xi$	$(s\cdot P)(s\cdot\Psi)/s^2$
S_6	$1/2$	$(s\cdot P)\,\Phi/s^2$
S_7, S_8, S_9	$-\frac{\xi}{2\bar{\xi}},\ \frac{\xi}{2},\ -\frac{1}{2}$	$\mathcal{S}_7,\ \mathcal{S}_8,\ \mathcal{S}_9$ as above

8. The result

With $q=\bar{\xi}k$, $D_0 = s^2/\bar{\xi}$, $M^2 = (\xi/\bar{\xi})s^2$, $C_{\rm UV} = P_{gg}/2N_c$:

$$\mathcal{W} \;=\; \int d^2r\; e^{iq\cdot r}\,\frac{T(r)}{D} \;=\; \mathcal{W}_{\rm UV} + \mathcal{W}_{\rm fin} + \mathcal{W}_{\rm reg},$$

$$\textbf{(I) UV log:}\qquad \mathcal{W}_{\rm UV} = C_{\rm UV}(\xi)\,(s\cdot P)\,\frac{\pi\bar{\xi}}{s^2}\ \log\frac{s^2}{\bar{\xi}\,m^2},$$

$$\textbf{(II) its finite companion:}\qquad \mathcal{W}_{\rm fin} = C_{\rm UV}(\xi)\,(s\cdot P)\,\frac{\bar{\xi}}{s^2}\left[\,\pi\log\frac{4e^{-2\gamma_E}\,\bar{\xi}}{q^2s^2} \;+\; 2\,s\!\cdot\!\Psi \;-\; \Phi\,\right],$$

$$\textbf{(III) the other terms:}\qquad \begin{aligned}\mathcal{W}_{\rm reg} = 2\xi\bar{\xi}\;P_is_j\,\mathcal{T}^{ij} \;+\; \left(\xi^2-\tfrac{5}{2}\xi+2-\tfrac{1}{\xi}\right)P\!\cdot\!\Psi \;+\; \frac{\bar{\xi}^{\,2}}{2\xi\,s^2}\,P_is_j\,\mathcal{N}^{ij}\\ -\;\frac{(s\cdot P)\,(s\!\cdot\!\Psi)}{2\xi\,s^2} \;+\; \frac{(s\cdot P)\,\Phi}{2s^2} \;-\; \frac{\xi}{2\bar{\xi}}\,\mathcal{S}_7 \;+\; \frac{\xi}{2}\,\mathcal{S}_8 \;-\; \frac{1}{2}\,\mathcal{S}_9.\end{aligned}$$

Group (I) is the only place m (or $1/\epsilon_{\rm UV}$) appears. Groups (II) and (III) are finite, and are given entirely by the six masters of §5 — one closed form (Φ) and five single α–integrals of Bessel functions. Nothing has been expanded, approximated or discarded anywhere between §2 and here.

9. Back to the cross section

Inserting into (1.11) — the bracket is T/P^2, so the z–integral of the bracket is $\mathcal{W}/P^2$ —

$$\frac{d^3N_2^{\rm NLO}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^4k^+}\int_{x,y,x',w'} e^{-ik\cdot w'}e^{ik\cdot x}\int_{\Lambda/k^+}^{1-\Lambda/k^+}\frac{d\xi}{2\pi}\,\frac{1}{\bar{\xi}}\;\frac{\mathcal{W}_{\rm UV}+\mathcal{W}_{\rm fin}+\mathcal{W}_{\rm reg}}{P^2}\;\big[\cdots\big]\rho^{b'}(x')\rho^{f}(y),$$

where $[\cdots]$ is the colour structure of (1.1). For group (I) the explicit $\bar{\xi}$ of $\mathcal{W}_{\rm UV}$ cancels the $1/\bar{\xi}$ exactly and one is left with a product of two Weizsäcker–Williams kernels times a logarithm of the dipole size:

$$\frac{1}{\bar{\xi}}\,\frac{\mathcal{W}_{\rm UV}}{P^2} = \pi\,\frac{P_{gg}(\xi)}{2N_c}\;\frac{(x'-w')\cdot(y-x)}{(x'-w')^2\,(y-x)^2}\;\log\frac{(y-x)^2}{\bar{\xi}\,m^2},$$

which is the object that pairs with the $-2/\epsilon$ of $A^{(3)}_{N_c}$ and becomes $\alpha_s\big((y-x)^{-2}\big)$. Groups (II) and (III) carry a different transverse structure — they are *not* proportional to the WW×WW kernel — and this is precisely the content that was missing before.

10. The p^+ integration

10.1 Group (I): P_{gg}, and b_0

Write $dp^+ = k^+d\xi,\ \lambda = \Lambda/k^+,\ \ell_k = \log(k^+/\Lambda),\ L = \log\big[(y-x)^2/m^2\big]$; the logarithm of §9 is $L + \log\frac{1}{1-\xi}$. All six integrals are elementary:

piece of $P_{gg}/2N_c$	**from**	$\int_\lambda^{1-\lambda} d\xi$	$\int_\lambda^{1-\lambda} d\xi\ \log\frac{1}{1-\xi}$
$\xi(1-\xi)$	Θ_1 (instantaneous)	$\frac{1}{6}$	$\frac{5}{36}$
$\dfrac{\xi}{1-\xi}$	Θ_2	$\ell_k - 1$	$\frac{1}{2}\ell_k^2 - 1$
$\dfrac{1-\xi}{\xi}$	Θ_4	$\ell_k - 1$	$\frac{\pi^2}{6} - 1$
total		$2\ell_k - \frac{11}{6}$	$\frac{1}{2}\ell_k^2 + \frac{\pi^2}{6} - \frac{67}{36}$

$$\int_\Lambda^{k^+-\Lambda} \frac{dp^+}{2\pi}\, \frac{P_{gg}(\xi)}{2N_c}\, \log\frac{k^+(y-x)^2}{(k^+-p^+)m^2} = \frac{k^+}{2\pi}\left[\left(2\ell_k - \frac{11}{6}\right)\log\frac{(y-x)^2}{m^2} + \frac{\ell_k^2}{2} + \frac{\pi^2}{6} - \frac{67}{36}\right].$$

Three readings. The double logarithm comes only from Θ_2 and the $\pi^2/6$ only from Θ_4; it is the instantaneous Θ_1 that turns -2 into $-\frac{11}{6}$ and -2 into $-\frac{67}{36}$. And $-\frac{11}{6} \times 2N_c = -\frac{11N_c}{3} = -b_0$ at $N_f = 0$: this row supplies, on its own, the $\frac{11N_c}{6}\delta(1-\xi)$ of the regularised splitting function, hence part of the running coupling.

If the p^+/k^+ **of (1.1) is kept** (see §1(ii)), every weight acquires an extra ξ and the plain integrals become $\int \xi \cdot \frac{1-\xi}{\xi} = \frac{1}{2}$, $\int \xi \cdot \frac{\xi}{1-\xi} = \ell_k - \frac{3}{2}$, $\int \xi \cdot \xi(1-\xi) = \frac{1}{12}$, total $\ell_k - \frac{11}{12}$ — one rapidity logarithm instead of two, and no b_0. Which convention is intended should be settled before this row is used numerically.

10.2 Groups (II) and (III): why their ξ–integral is not elementary

$q = \bar\xi k$ depends on ξ, and so do $M^2 = (\xi/\bar\xi)s^2$ and $D_0 = s^2/\bar\xi$. The masters therefore depend on ξ both through their coefficients and through their arguments, and $\int d\xi$ cannot be done in closed form. What *can* be established is the endpoint behaviour, which is what decides the rapidity logarithms.

10.3 $\xi \to 0$: a $z \to y$ logarithm that cancels between (II) and (III)

As $\xi \to 0$, $M^2 \to 0$ and D develops a genuine singularity at $r = s$. For any smooth f, $\int d^2r\, f(r)/[(r-s)^2 + M^2] = \pi f(s)\log(1/M^2) + \ldots$, so

$$\Psi^i\Big|_{\log} = \pi e^{iq\cdot s}\frac{s^i}{s^2}\Lambda_M, \quad \mathcal{N}^{ij}\Big|_{\log} = \pi e^{iq\cdot s}\frac{s^i s^j}{s^2}\Lambda_M, \quad \Phi\Big|_{\log} = \pi e^{iq\cdot s}\Lambda_M, \quad \mathcal{T}^{ij}\Big|_{\log} = \pi e^{iq\cdot s}\left[\tfrac{s^i s^j}{s^4} - \tfrac{\delta^{ij}}{2s^2}\right]\Lambda_M,$$

with $\Lambda_M \equiv \log(1/M^2) = \log(\bar\xi/\xi s^2)$. For $\mathcal{S}_{7,8,9}$ the numerators vanish at $r = s$, and the logarithm comes only from the $e^{iq\cdot\rho}$ expansion, $\mathcal{S}_k|_{\log} = i\pi e^{iq\cdot s}(q\cdot c_k)\Lambda_M/2s^2$ with $c_7 = (s\cdot P)s$, $c_8 = (P\cdot s)s$, $c_9 = s^2 P$ — all $O(1)$, not $O(1/\xi)$. Keeping only the $1/\xi$ poles ($c_3 \to -\frac{1}{\xi}$, $c_4 \to \frac{1}{2\xi}$, $c_5 \to -\frac{1}{2\xi}$; sum $-\frac{1}{\xi}$):

$$\mathcal{W}_{\text{reg}} \xrightarrow[\xi\to 0]{} -\frac{\pi}{\xi}\, e^{iq\cdot s}\, \frac{s\cdot P}{s^2}\, \Lambda_M + O\!\left(\frac{1}{\xi}\right), \qquad \mathcal{W}_{\text{fin}} \xrightarrow[\xi\to 0]{} +\frac{\pi}{\xi}\, e^{iq\cdot s}\, \frac{s\cdot P}{s^2}\, \Lambda_M + O\!\left(\frac{1}{\xi}\right),$$

because $C_{\text{UV}} \to 1/\xi$, $D_0 \to s^2$ and $(2s\cdot\Psi - \Phi)|_{\log} = (2-1)\pi e^{iq\cdot s}\Lambda_M$. The two cancel exactly. Both statements are confirmed numerically to five digits (§12E), where $\xi(\mathcal{W}_{\text{fin}} + \mathcal{W}_{\text{reg}})$ is seen to approach a constant while each piece separately grows like $\log(1/\xi)$.

> **Consequence.** The $z \to y$ (collinear) logarithm is an artefact of where the UV cut is placed; it is absent from the sum. What survives at $\xi \to 0$ is a simple pole $\propto 1/\xi$, i.e. *one* rapidity logarithm ℓ_k from the finite and regular terms together — in addition to the ones carried by P_{gg} in group (I).

10.4 $\xi \to 1$: the regular block does not vanish

At fixed q, as $\bar\xi \to 0$ one has $M^2 \to \infty$: every master falls off like $1/M^2$ and the only survivor is $-\frac{\xi}{2\bar\xi}\mathcal{S}_7 \to -\frac{\xi}{2\bar\xi}\cdot\frac{(s\cdot P)A_7}{M^2}$, so

$$\lim_{\xi\to 1} \mathcal{W}_{\text{reg}}\Big|_{q\text{ fixed}} = -\frac{(s\cdot P)}{2s^2}\, A_7(q,s) \qquad \text{(verified to 6 digits, §12F).}$$

With the physical $q = \bar\xi k$ the two limits do not commute — $A_7 \sim 2\pi\log(1/qs)$ as $q \to 0$ and the other masters grow too — but the conclusion is the same and is what matters: $\mathcal{W}_{\text{reg}}$ tends to a **finite nonzero constant** at the endpoint (numerically $+0.868\ldots$ for the test kinematics), so the explicit $1/\bar\xi$ of the cross section turns it into a rapidity logarithm ℓ_k with a coefficient that is *not* contained in P_{gg}. Any rapidity subtraction applied to this row must therefore act on the regular terms as well, not only on group (I).

11. What the split does and does not mean

- **Unambiguous.** The coefficient of $\log(1/m^2)$ — equivalently of $1/\epsilon_{\rm UV}$ — is $C_{\rm UV}(\xi)(s\cdot P)\pi\bar{\xi}/s^2$, i.e. $P_{gg}/2N_c$ times two WW kernels. So is the algebraic identity of §4, and so is the sum $\mathcal{W}_{\rm UV}+\mathcal{W}_{\rm fin}+\mathcal{W}_{\rm reg}$.
- **Scheme–dependent.** The boundary between (I) and (II) is the argument chosen for the logarithm, here $D_0=s^2/\bar{\xi}$. Choosing q^{-2} or s^2 instead moves constants (and, at $\xi\to 1$, a $\log\bar{\xi}$) between the two. The boundary between (II) and (III) is fixed by the traceless projection of §4, which is natural but not forced.
- **The $U(z)$ caveat.** In (1.1) the colour structure contains $U^{ce}(z)$, so the z–integral is not literally separable from it; (1.11) writes it as if it were. For group (I) this is harmless and is in fact the point: the divergence is local, $z\to x$, so $U^{ce}(z)\to U^{ce}(x)$ there and the coefficient multiplies a genuinely local colour structure — which is why it can be absorbed into the running coupling. For groups (II) and (III) the results above are the z–integral *at fixed colour structure*, i.e. they are exact if $U^{ce}(z)$ is taken outside, and otherwise they are the decomposition of the *z–integrand*, to be convolved with $U^{ce}(z)$. The algebraic split of §4 is unaffected either way, since it is pointwise in r.
- **$d_\perp$.** Only c_1 (hence the $\xi(1-\xi)$ of P_{gg}) knows about the transverse dimension — but only apparently. Continuing the transverse plane to $d_\perp$ dimensions changes the angular average as well, $\langle\hat{r}^i\hat{r}^j\rangle=\delta^{ij}/d_\perp$, and the two factors cancel: $c_1\langle S_1\rangle=\frac{d_\perp}{d_\perp}\xi\bar{\xi}\frac{s\cdot P}{r^2}$. The $\xi(1-\xi)$ piece of P_{gg} therefore receives no $O(\epsilon)$ correction from this term.

12. Numerical verification

Everything above is checked. The algebra is checked symbolically (sympy) directly against Eq. (1.1); the integrals are checked against direct two–dimensional quadrature of $\int d^2r\, e^{iq\cdot r}F(r)$, done in polar coordinates with q rotated onto the x–axis: periodic–trapezoid in the angle, adaptive quadrature in the radius out to $R_n=R_0+n\pi/q$, and repeated averaging of the partial sums to kill the slowly decaying oscillatory tail.

A. the algebra (sympy, exact)

```
Theta_1 == (1.4)                        : True
Theta_2 == (1.5)                        : True
Theta_3 == (1.6)/(1.7)                  : True
Theta_4 == (1.8) as printed             : False      <-- the factor-2 typo
Theta_4 == (1.8) with s^2 -> 2 s^2      : True
Theta_5 == (1.9)                        : True
Theta_6 == (1.10)                       : True
T (d_perp=2)  ==  sum_k c_k S_k         : True
T - C_UV (s.P)/r^2  ==  T_fin           : True
```

B–G. the integrals

```
kinematics:  s = [ 0.7 -0.4]  q = [0.9 0.6]  P = [1.3 0.5]  xi = 0.3
             s^2 = 0.6500  M^2 = 0.278571  D0 = 0.928571  s.P = 0.7100  q.s = 0.3900

A. master integrals:  alpha-representation  vs  direct 2D quadrature
           alpha-representation                       2D quadrature                              rel.diff
Psi^0      +2.7273301853+3.3432229840j                +2.7273301436+3.3432230085j                1.1e-08
Psi^1      -2.1029242522+1.3184267910j                -2.1029242881+1.3184267937j                1.4e-08
N^00       +2.0493314046+2.0734030364j                +2.0493311443+2.0734031386j                9.6e-08
N^01       -1.9056399937-0.3615375386j                -1.9056401726-0.3615376223j                1.0e-07
N^11       +2.6969517015-0.1224200369j                +2.6969516329-0.1224201107j                3.7e-08
Tt^00      +0.2544265956+0.9149478158j                +0.2544265787+0.9149478226j                1.9e-08
Tt^01      -1.4224659992-0.1649205388j                -1.4224660382-0.1649205438j                2.7e-08
A7         +2.4533167153+0.4845541087j                +2.4533161332+0.4845547275j                3.4e-08
A8         -0.3592796084-1.4936545930j                -0.3592788373-1.4936536083j                8.1e-07
A9         -0.9001711373+1.2449007205j                -0.9007825997+1.2446497402j                4.3e-04

B. exact identities
   trace N^ij - Phi                      : 4.58e-15
   trace Tt^ij                           : 1.39e-15
   G(m) alpha-rep vs (2pi/D0)K0(mq)+(2s.Psi-Phi)/D0, m=1e-3 : 5.60e-07
   A8(P=s) - A9(P=s)                     : 0.00e+00

C. the regular block  W_reg  (assembled from masters)  vs direct quadrature of T_fin/D
   xi=0.30  assembled -4.48077798-10.60311732j                 quad -4.47731510-10.60169400j                  rel
3.3e-04
   xi=0.65  assembled -0.48091094-2.99952516j                  quad -0.48083411-2.99949381j                   rel
2.7e-05

D. the UV coefficient: d W / d log(1/m^2) of the *regulated original* integrand
   xi=0.30  measured +7.137881   predicted C_UV (s.P) pi/D0 = +7.138842   rel 1.3e-04
   xi=0.65  measured +3.150126   predicted C_UV (s.P) pi/D0 = +3.150493   rel 1.2e-04

E. xi -> 0 : the ln(1/M^2) of the companion cancels that of the regular block
   xi        xi*W_companion            xi*W_reg                  xi*(W_comp+W_reg)
   1e-02     +10.6714+5.5627j          -13.1341-9.2589j          -2.462678-3.696233j
   1e-03     +18.1362+8.6060j          -20.6686-12.3818j         -2.532414-3.775854j
   1e-04     +25.4732+11.6164j         -28.0125-15.4003j         -2.539311-3.783930j
   1e-05     +32.7856+14.6213j         -35.3256-18.4061j         -2.539999-3.784738j
   1e-06     +40.0943+17.6255j         -42.6344-21.4103j         -2.540068-3.784819j
   coefficient of ln(1/M^2)/xi in W_reg: predicted -3.173905-1.304649j
                                         measured  -3.174994-1.305007j   ratio 1.00033

F. xi -> 1 at fixed q :  W_reg -> -(s.P) A7 /(2 s^2)
   prediction -1.3398884-0.2646411j
   xibar=1e-03    W_reg = -1.3394803-0.2763443j
   xibar=1e-04    W_reg = -1.3398468-0.2658110j
   xibar=1e-05    W_reg = -1.3398842-0.2647581j

G. the p+ integrals of P_gg/(2Nc)   (mpmath, 30 digits, lambda=1e-8)
   sum of plain integrals   35.00802817457   2 l_k - 11/6      = 35.00802815457
   sum of log-weighted ones 169.4445626753   l_k^2/2+pi^2/6-67/36 = 169.444562491
```

Comments on the table. A_9 is the slowest–converging quadrature (its integrand falls only as r^{-2} with a nonzero angular average); pushing the averaging depth from 14 to 20 levels moves the quadrature from -0.90598 to -0.90078 against the α–representation's -0.90017, i.e. it is converging onto it. Row D measures $d\mathcal{W}/d\log(1/m^2)$ on the *original* integrand with $r^2 \to r^2+m^2$ everywhere — a regularisation different from the one used in §6, which is the point: the log coefficient does not care.

13. Summary card

Integrand $T(r) = C_{\rm UV}(\xi)\dfrac{s\cdot P}{r^2} + T_{\rm fin}(r)$, $\;C_{\rm UV} = \xi\bar\xi + \dfrac{\xi}{\bar\xi} + \dfrac{\bar\xi}{\xi} = \dfrac{P_{gg}}{2N_c}$ (exact, pointwise).

z–integral $\mathcal{W} = \mathcal{W}_{\rm UV} + \mathcal{W}_{\rm fin} + \mathcal{W}_{\rm reg}$ of §8, built from Φ, Ψ^i, $\mathcal{N}^{ij}$, $\mathcal{T}^{ij}$, $A_{7,8,9}$ and $\mathcal{S}_{7,8,9}$.

UV $\dfrac{1}{\bar\xi}\dfrac{\mathcal{W}_{\rm UV}}{P^2} = \pi\dfrac{P_{gg}(\xi)}{2N_c}\dfrac{(x'-w')\cdot(y-x)}{(x'-w')^2(y-x)^2}\log\dfrac{(y-x)^2}{\bar\xi m^2}$, $\;\displaystyle\int\frac{dp^+}{2\pi} \to \frac{k^+}{2\pi}\Big[(2\ell_k - \tfrac{11}{6})L + \tfrac{1}{2}\ell_k^2 + \tfrac{\pi^2}{6} - \tfrac{67}{36}\Big].$

Everything else finite; carries a different transverse structure; has its own $1/\xi$ and $1/\bar\xi$ endpoint behaviour, hence its own rapidity logarithms (§10.3–10.4); and its $\log M^2$ cancels between groups (II) and (III).

Code

`zint/src/algebra.py` — sympy reduction of Eq. (1.1) to $T(r)$, the nine–structure basis and the UV split (§12A). `zint/src/masters.py` — the α–representations of §5. `zint/src/quad2d.py` — the oscillatory 2D quadrature. `zint/src/assemble.py` — the assembly of §8. `zint/src/run_all.py` — every number in §12, written to `zint/RESULTS.txt`.